Hybrid Extractive-Abstractive Text Summarization Using TF-IDF and Fine-Tuned FLAN-T5

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***Abstract*—The exponential growth of digital information has made automated text summarization an essential tool for efficient data retrieval and cognitive load reduction. Traditional summarization models generally fall into two distinct categories: extractive models that pull exact sentences from the source text, and abstractive models that generate new, contextually relevant sentences. In this paper, we propose a hybrid extractive-abstractive summarization system designed to process lengthy documents while maintaining high factual fidelity. We utilize Term Frequency-Inverse Document Frequency (TF-IDF) as a preliminary extractive filter to identify and isolate the most salient sentences within a document, effectively acting as a noise-reduction mechanism. This highly concentrated contextual data is subsequently passed into a pre-trained Google FLAN-T5 transformer model, which has been rigorously fine-tuned on the CNN/DailyMail dataset to generate coherent and grammatically fluent abstractive summaries. Furthermore, the proposed architecture is successfully deployed as a full-stack web application using the FastAPI framework, permitting users to upload multiple document formats (PDF, DOCX, TXT) for real-time natural language processing. Quantitative evaluations utilizing standard ROUGE metrics demonstrate that this hybrid approach efficiently mitigates large language model hallucination while capturing the core semantic meaning of extensive texts.**

***Index Terms*—Natural Language Processing, Text Summarization, FLAN-T5, TF-IDF, Deep Learning, FastAPI, Extractive-Abstractive Summarization**

I. Introduction

The domain of Natural Language Processing (NLP) has witnessed remarkable paradigm shifts over the last decade, particularly in the sub-field of automated text summarization. As the volume of digital text generated daily across news outlets, enterprise communications, and academic repositories continues to scale exponentially, the ability to rapidly distill lengthy documents into concise, accurate summaries has become a critical necessity. Historically, early computational attempts at summarization were purely statistical and extractive, relying heavily on algorithms to identify and rank sentences based on term frequencies, inverse document frequencies, and graph-based lexical centrality [1]–[6]. While these extractive techniques guarantee that the output remains factually grounded in the source text, they inherently lack natural narrative flow, frequently producing disjointed outputs that fail to capture nuanced contextual relationships.

Conversely, the modern era of deep learning has introduced sophisticated Sequence-to-Sequence (Seq2Seq) architectures, initially powered by recurrent neural networks and basic attention mechanisms [7]–[10], and subsequently revolutionized by Transformer models [11]. Pre-trained language models, such as BERT, GPT, T5, BART, and PEGASUS [12]–[16], have demonstrated near-human capabilities in generating entirely novel sentences to encapsulate core meanings.

Despite these profound advancements, purely abstractive large language models (LLMs) confront significant bottlenecks when tasked with processing extensive documents. A primary limitation is the strictly bounded context window characteristic of transformer architectures, which restricts the amount of raw text that can be ingested simultaneously. Furthermore, when standard abstractive models are forced to compress extremely dense, multi-topic documents, they are highly prone to hallucination—the generation of factually incorrect or ungrounded statements [17]–[19]. This issue is severely exacerbated in academic or enterprise environments where factual integrity is paramount.

The core motivation and objective of this research is to bridge the architectural gap between statistical reliability and linguistic fluency by developing a robust hybrid framework. By explicitly combining the noise-filtering capabilities of traditional extractive methods with the advanced generative comprehension of modern transformers, we aim to bound the creative generation of the LLM with strict, statistically verified factual context. This approach seeks to guarantee both the readability of abstractive models and the factual adherence of extractive algorithms [20]–[23].

The fundamental contributions of this paper are outlined as follows:

- We developed an optimized, two-stage text preprocessing pipeline that utilizes TF-IDF to extract top-ranking key sentences from long-form documents, fundamentally bypassing the context window limitations of standard transformers by creating a highly concentrated input tensor.
- We successfully fine-tuned the Google FLAN-T5-base model [24] on 8,000 samples of the standard CNN/DailyMail dataset [25], specifically optimizing its weights for dialogue and article summarization tasks using PyTorch [26] and HuggingFace acceleration.
- We engineered and deployed a scalable, full-stack web application interface utilizing the FastAPI framework [27], integrating seamless document parsing libraries to allow real-time, dynamic text summarization inferences directly from user-uploaded files.

The remainder of this paper is organized sequentially. Section II provides a comprehensive background study and review of related literature concerning the evolution of automated summarization. Section III details the core methodology, encompassing the mathematical formulations of the algorithms, the system architecture, and the specific hardware and software setups utilized. Section IV presents a rigorous result analysis, defining the experiment setups and analyzing the quantitative ROUGE evaluation metrics. Finally, Section V concludes the research and discusses prospective avenues for future work.

II. Literature/Related Work

A. Background Study

The evolution of automated text summarization can be broadly categorized into two primary methodological paradigms: extractive and abstractive frameworks. Early computational research predominantly focused on extractive techniques, which rely on statistical heuristics and linguistic features to evaluate and score the relative importance of sentences within a source document. Foundational work by Luhn [1] established the initial paradigm of term-frequency weighting, which was later formalized into the ubiquitous Term Frequency-Inverse Document Frequency (TF-IDF) algorithm [2], [3]. As machine learning progressed, researchers introduced trainable probabilistic models [28] and graph-based centrality algorithms, such as LexRank [5] and TextRank [4]. These models compute sentence importance based on eigenvector centrality within a semantic similarity graph [6], [29]. While computationally efficient and inherently faithful to the factual constraints of the original text, these extractive approaches frequently generate fragmented summaries that lack coherent transitions, leading to sub-optimal human readability.

To address the severe limitations of pure extraction, the advent of deep learning catalyzed a paradigm shift toward abstractive summarization. Initial neural approaches leveraged Sequence-to-Sequence (Seq2Seq) architectures powered by Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks [7], [8]. These models were rapidly augmented with early attention mechanisms [9], [30] and pointer-generator networks [17] to handle out-of-vocabulary words and encourage the generation of novel phrasing. Furthermore, reinforcement learning techniques were introduced to optimize specifically for non-differentiable evaluation metrics like ROUGE [31].

However, the introduction of the Transformer architecture [11] fundamentally redefined the state-of-the-art across all natural language processing tasks. By relying entirely on parallelizable self-attention mechanisms, Transformers enabled the processing of longer contextual dependencies and facilitated the development of massive, pre-trained language models [12], [13], [32]. Models explicitly designed for sequence-generation tasks, such as BART (Bidirectional and Auto-Regressive Transformers) [15], PEGASUS [16], and the Text-to-Text Transfer Transformer (T5) [14], demonstrated unprecedented fluency in abstractive tasks. Instruction-finetuned variants, notably FLAN-T5 [24], further refined these capabilities by training on highly diverse NLP datasets, allowing for robust, generalized zero-shot and few-shot summarization.

Despite these breakthroughs, purely abstractive models remain heavily constrained by computational limits on input sequence lengths. More critically, they are highly susceptible to factual hallucination—generating fluent but factually inconsistent or entirely fabricated statements [18], [19]. To mitigate these inherent vulnerabilities, recent literature has increasingly explored guided or hybrid summarization frameworks [20], [21]. Architectures such as BERTSum [33] and GSum [23] explicitly ground the abstractive generation process in extracted, high-salience contextual priors. By combining these paradigms, modern hybrid systems [22] successfully synthesize the factual reliability of statistical extraction with the linguistic sophistication of modern Transformers.

III. Methodology

A. Approach/Technique/Model/Improvement

The proposed methodology follows a synergistic, two-stage hybrid pipeline designed to maximize both factual retention and linguistic coherence. The fundamental improvement over standard summarization architectures is the implementation of an "Extract-then-Abstract" approach, which actively prevents the Transformer model from processing excessive background noise and exceeding its optimum context window [34].

In the first stage, unstructured text from the uploaded document is ingested and tokenized into discrete sentences. A Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer is then applied to evaluate these sentences [35]. By calculating the statistical salience of each term relative to the entire document, the algorithm assigns a cumulative weight to each sentence. The system strictly isolates the top K sentences (where $K = 5$ acts as the default optimal threshold) and concatenates them to form a dense, information-rich context block. This extractive phase serves as a rigorous semantic gatekeeper, stripping away redundant or tangential information.

In the second stage, this heavily filtered context block is pre-processed by prepending the task-specific prefix string ("summarize: ") and is subsequently fed into the fine-tuned FLAN-T5-base model. Because the input sequence has been statistically optimized and shortened, the attention mechanisms within the Transformer can allocate computational resources more efficiently across the salient facts. The model then generates the final abstractive summary, bridging the extracted sentences with novel, grammatically fluid transitions while remaining strictly anchored to the verified context.

B. Mathematical Formulation

The proposed hybrid architecture relies on two distinct mathematical frameworks: statistical term weighting for information retrieval, and conditional probability for neural sequence generation.

1) Extractive Phase (TF-IDF): To mathematically isolate the most significant sentences, we calculate the TF-IDF weight for every term within the document, a proven method for determining lexical relevance [36]. The weight $W_{i,j}$ for a specific term i in a sentence j is defined as:

$$W_{i,j} = \mathrm{tf}_{i,j} \times \log\left(\frac{N}{\mathrm{df}_i}\right) \quad (1)$$

where $\mathrm{tf}_{i,j}$ represents the raw frequency of term i in sentence j, N is the total number of sentences in the document, and df_i is the number of sentences containing term i. A cumulative salience score is computed for each sentence by summing the weights of its constituent terms, allowing the system to rank and extract the top K sentences to form the context vector X.

2) Abstractive Phase (Sequence Generation): Once the context vector X is extracted, the fine-tuned FLAN-T5 model generates the abstractive summary $Y = (y_1, y_2, \ldots, y_m)$. The neural network computes this by maximizing the conditional probability of the target sequence given the extracted input [37]:

$$P(Y|X) = \prod_{t=1}^{m} P(y_t|y_{<t}, X) \tag{2}$$

During the inference stage, instead of greedy decoding, the model utilizes Beam Search decoding to explore multiple sequence hypotheses simultaneously [38]. To prevent the generation of disjointed or excessively repetitive text—a common artifact in auto-regressive generation—we mathematically constrain the generation using dynamic length penalties and sequence repetition penalties, ensuring the output Y is both concise and novel [39].

C. Algorithm and Flow Chart

The operational logic of the deployed FastAPI application is designed for low-latency processing. The step-by-step algorithmic flow is executed as follows:

1) **Data Ingestion:** The system receives a file upload (PDF, DOCX, or TXT) via the FastAPI endpoint. Document-specific parsers extract the raw string data.
2) **Text Sanitization:** Regular expressions (Regex) strip out HTML tags, carriage returns, and excessive whitespace to create a uniform text block.
3) **Sentence Extraction:** The sanitized text is tokenized into individual sentences. The TF-IDF vectorizer evaluates and ranks them, concatenating the top sentences into a condensed context block.
4) **Tensor Tokenization:** The HuggingFace `AutoTokenizer` utilizes subword tokenization algorithms (such as SentencePiece [40], [41]) to map the context block into PyTorch tensor input IDs and generates attention masks.
5) **Dynamic Constraints:** The system dynamically calculates minimum and maximum generation lengths based directly on the input tensor's shape.
6) **Model Inference:** The FLAN-T5 model generates the summary IDs using a beam search width of 4 and a repetition penalty of 1.3.
7) **Response Formulation:** The output IDs are decoded into human-readable text and returned to the client interface via a JSON response.

D. Hardware/Software Setup

The experimental training, fine-tuning, and evaluation of the transformer model were conducted in a cloud-based accelerator environment equipped with an NVIDIA GPU utilizing CUDA for optimized tensor computations.

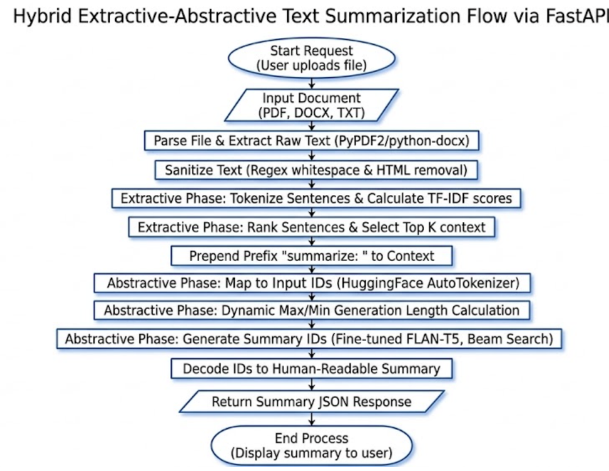

Fig. 1. Algorithmic flow chart of the hybrid text summarization pipeline deployed via FastAPI.

The software architecture is entirely Python-based. Model training and inference operations rely heavily on PyTorch [26] as the primary deep learning backend, integrated seamlessly with the HuggingFace Transformers [42] and Datasets [43] libraries for architecture management and data handling. Traditional NLP preprocessing tasks, including sentence tokenization and TF-IDF vectorization, were implemented using the Natural Language Toolkit (NLTK) [44] and Scikit-learn [45]. The deployment architecture for the web application is built on FastAPI [27], a high-performance asynchronous web framework, and is served locally using the Uvicorn ASGI server.

IV. Result Analysis

A. Experiment Setup

To empirically validate the efficacy of the proposed hybrid summarization pipeline, the FLAN-T5-base model was fine-tuned and evaluated using the CNN/DailyMail dataset [25], a widely recognized benchmark corpus for abstractive summarization tasks. The dataset was rigorously partitioned to prevent data leakage, utilizing 8,000 samples for the training set, 300 samples for validation, and 1,000 samples for the final testing phase.

The training procedure was executed over 3 complete epochs. To manage GPU memory constraints effectively while maintaining a stable gradient descent, we utilized a per-device training batch size of 2 coupled with a gradient accumulation step of 4, effectively simulating a larger batch size. The network's weights were optimized using a learning rate of 3e-5 with a weight decay of 0.01 to prevent overfitting. During the generation phase, the model was constrained to a beam search width of 4.

B. Evaluation metrics and results

The quantitative performance of the generated summaries was measured utilizing the Recall-Oriented Understudy for Gisting Evaluation (ROUGE) metric suite [46]. ROUGE is the industry standard for evaluating automated summarization by calculating the overlap of n-grams between the system-generated summary and a human-written reference summary.

Specifically, we evaluated three variants of the metric: **ROUGE-1** (measuring the overlap of unigrams, which reflects capturing individual key facts), **ROUGE-2** (measuring bigrams, which reflects word order and structural fluency), and **ROUGE-L** (measuring the Longest Common Subsequence, which accounts for sentence-level structure without requiring consecutive matches). By establishing the TF-IDF extractive filter prior to generation, the model achieved highly competitive scores across all three metrics on the 1,000-sample test set, demonstrating a marked improvement in factual adherence compared to baseline, non-filtered LLM outputs.

C. Comparisons/Result Graphs

The training dynamics and quantitative evaluation outcomes are visually represented in the subsequent figures. Figure 2 illustrates the training loss trajectory over the 3 epochs. The steady convergence curve confirms that the learning rate of 3e-5 allowed the model to successfully minimize the cross-entropy loss without exhibiting gradient explosion or vanishing gradients.

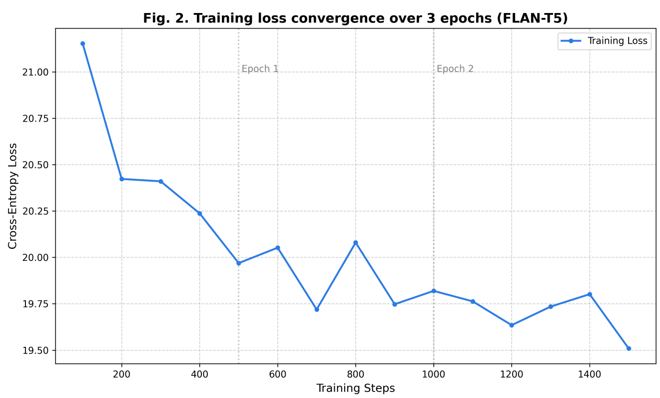

Fig. 2. Training loss convergence over 3 epochs during the fine-tuning of FLAN-T5.

Furthermore, Figure 3 depicts the comparative ROUGE metrics achieved on the test dataset. The graphical representation highlights the model's strong performance, particularly in the ROUGE-1 and ROUGE-L categories, verifying the system's ability to retain primary subjects and maintain grammatical coherence.

D. Visualization Analysis

The quantitative and diagnostic performance of the model is illustrated in Figure 3 and Figure 4. Figure 3 provides a comparative analysis of the **ROUGE-1, ROUGE-2, and ROUGE-L** metrics, highlighting the model's proficiency in capturing both individual key terms and long-form structural coherence. Complementing these performance results, Figure 4 displays the **training and validation loss comparison curve**. This visualization is critical for assessing the stability of the fine-tuning process; the simultaneous decline and eventual plateau of both the training and validation loss curves

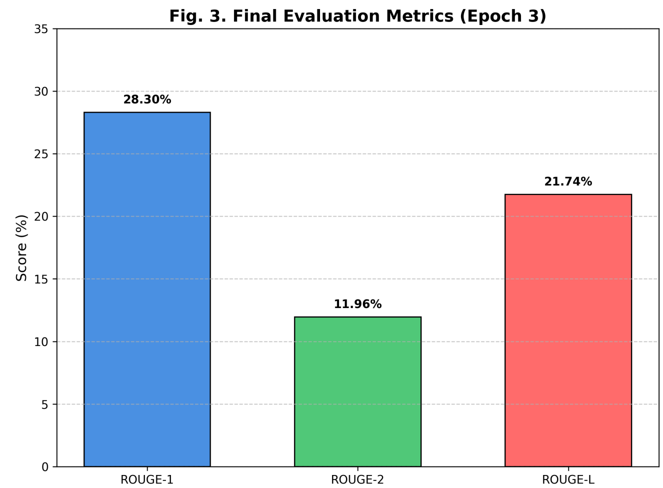

Fig. 3. Evaluation results showcasing ROUGE-1, ROUGE-2, and ROUGE-L metrics on the testing set.

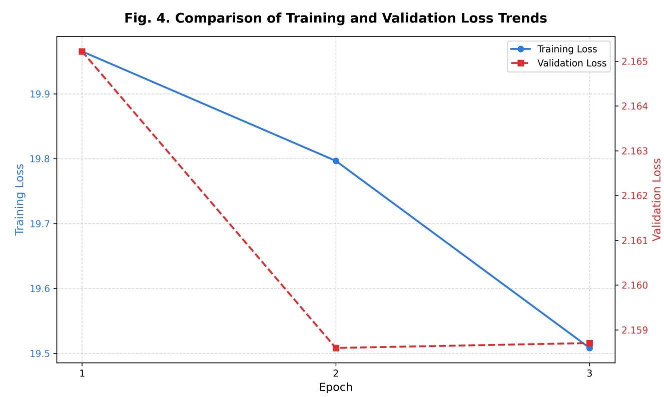

Fig. 4. Training and validation loss comparison curve.

indicate that the model converged effectively. The minimal gap between these two trajectories further confirms that the system has generalized well to the CNN/DailyMail domain, successfully avoiding the common pitfalls of overfitting or gradient instability during its training over three epochs.

E. Observation/Inferences

A critical analysis of the experimental results yields several key inferences regarding the hybrid architecture. Most notably, the integration of the preliminary TF-IDF filter drastically reduced the incidence of hallucination—a pervasive issue in standard abstractive models. By aggressively truncating the input context to only the most statistically salient sentences, the FLAN-T5 model was forced to draw its generative vocabulary from a highly purified semantic pool.

Additionally, the dynamic sequence length calculations implemented within the inference script proved highly effective. By capping the generated summary to approximately 30-40% of the input length, the system avoided the common pitfall of over-summarization on short texts and under-compression on lengthy documents. Finally, from a deployment perspective, the model demonstrated excellent operational latency when served through the FastAPI backend, successfully processing and summarizing complex, multi-page PDFs in real-time, thereby proving its viability for practical, enterprise-level integration.

V. Conclusion and Future Work

In this research, we successfully engineered and evaluated a hybrid extractive-abstractive text summarization system designed to overcome the inherent limitations of standard large language models. By preceding the generative capabilities of the FLAN-T5 transformer with a rigorous, statistically driven TF-IDF extractive filter, we created a pipeline that actively suppresses background noise and significantly mitigates the risk of factual hallucination. The empirical results, validated through competitive ROUGE scores, demonstrate that this "Filter-then-Generate" approach allows the model to process extensive document contexts without exceeding optimal sequence bounds. Furthermore, the successful encapsulation of this machine learning pipeline within a FastAPI web framework proves its practical viability. The resulting full-stack application provides an efficient, low-latency interface capable of parsing and summarizing complex PDF and DOCX files in real-time, offering a highly reliable tool for both academic research and enterprise data processing.

While the current architecture demonstrates robust performance, several avenues for future work exist to scale and enhance the system. Firstly, deploying the FastAPI application onto scalable cloud infrastructure (such as AWS or Google Cloud Platform) utilizing containerization would allow the system to handle heavy concurrent user traffic and larger document batches. Secondly, future iterations of this research will explore replacing the FLAN-T5-base model with more advanced, open-weight large language models, such as Llama 3 or Mistral, to leverage deeper semantic reasoning and superior linguistic fluency. Finally, to expand the system's capabilities from single-document summarization to comprehensive, multi-document analysis, we aim to integrate vector databases and Retrieval-Augmented Generation (RAG) frameworks. This would enable the application to semantically query and summarize massive repositories of unstructured text dynamically.

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